SET-C

15-05-2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025			
Maximum Marks: 50	Examination: End-Semester Examination		Duration: 2.5 Hrs.
Programme code: 05 Programme: Honors in Software and Mobile Application Development	Class: SY		Semester: IV (SVU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/Center: COMP		
Course Code: 216H05C401	Name of the Course: Fundamental of Swift		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q.1	Attempt any **four**	20
i)	What is the purpose of the `throws` keyword in Swift functions? Provide a throwing function.	05
ii)	What are variadic parameters? Illustrate their usage in a function.	05
iii)	What is a protocol in Swift? Write an example showing a class conforming to a protocol.	05
iv)	Write a Swift class that inherits from a generic class. Explain the use case.	05
v)	What is the `OptionSet` protocol in Swift? Describe with a use case. How does `OptionSet` differ from `enum` in Swift?	05
vi)	Describe the difference between `Decodable`, `Encodable`, and `Codable`.	05
Q.2	Attempt any **one**	
i)	**Write a swift program for the following scenario:** An event app tracks attendees and performs operations like checking for duplicates or preferences. **Requirements:** – Create a protocol `AttendeeInfo` with a method `getDetails()`. – Define a class `Attendee` conforming to this protocol with properties like `name`, `id`, and `eventPreference`. – Use a **Set** to store unique attendee IDs. – Write a function that accepts a closure to filter attendees based on their event preference.	10
ii)	**Write a swift program for the following scenario:** – Create an enum `OrderStatus` with cases: `Placed`, `Dispatched`, `InTransit`, and `Delivered`, each returning a message. – Define a struct `Order` with `id`, `status`, and optional `customerName`. – Write a `trackOrder(order:)` function using guard to validate the optional order and customer name. – Print a greeting and the order status message; demonstrate with valid,	10

	missing name, and nil cases.	
Q.3	Attempt the following	
i)	Using UIKit in Storyboard, create a **NewsLetter** registration form with the following fields: – Full Name (`UITextField`), Gender (`UISwitch`), Email (`UITextField`) and Age (`UISlider`) and a `UILabel` that dynamically displays the chosen age. – Add a Submit `UIButton` and a Newsletter Subscription (`UISwitch`) switch. - Use a vertical `UIStackView` to control the layout. - Add Appropriate Auto Layout constraints to to make the UI responsive across all screen sizes . - To display the entered details using `UILabel`, add a second screen and execute a navigation segue upon submission.	20